SIMULATION METHODS AND RESULTS

Team VectorCraft

WHAT WE DID:

- **Tool:** a Python model (NumPy, SciPy, Matplotlib).
- **It covers:** the sizing in, a one-axis simulation of the stand-off loop in wind, a motor-out thrust check, the magnetic-field estimate and an arm bending check.

MODEL AND INPUTS:

- **Stand-off model:** a point mass moving toward and away from the conductor. Tilt follows a second-order response ($\omega n = 9$ rad/s, $\zeta = 0.8$) to the tilt command and thrust rises to hold altitude ($T = W / \cos \theta$). Drag is $0.5 \cdot \rho \cdot CdA \cdot |v_{rel}| \cdot v_{rel}$ with the wind always blowing toward the line. The controller is a range loop from the LiDAR feeding a PI velocity loop.
- **Inputs:** AUW 5.68 kg, CdA 0.12 m^2, nominal range 4.0 m, LiDAR 40 Hz with 0.03 m noise and 50 ms latency, velocity noise 0.05 m/s, tilt limit 25°, max thrust 19.6 N per motor, time step 2.5 ms. Wind is a steady 8 m/s with small turbulence and a 1-cosine gust to 12 m/s over 6 s [11]. We start in equilibrium and ignore the first 20 s, so the results show the gust only.

Table 9. Gust results (nominal 4.0 m, hard floor 3.0 m)

Case	Closest range	Excursion	Peak tilt	Peak motor	Result
Ideal sensing	3.70 m	0.30 m	11.6°	9.5 N	Above floor
LiDAR noise and latency	3.68 m	0.32 m	11.7°	9.5 N	Above floor
Noise plus 0.5 s LiDAR dropout	3.69 m	0.31 m	12.0°	9.5 N	Above floor
Weak velocity loop (no integral)	2.57 m	1.43 m	11.8°	9.5 N	**Breaks the floor**

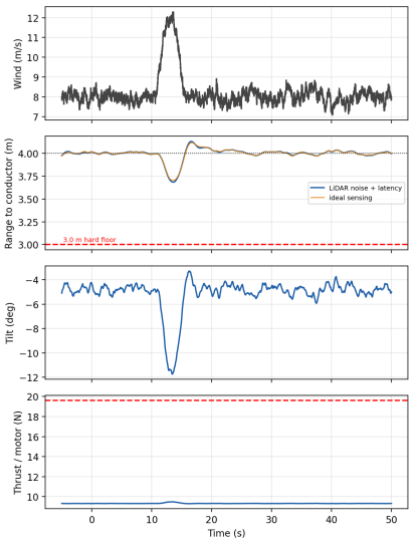


Figure 3. Wind, range to conductor, tilt and motor thrust in the gust case.

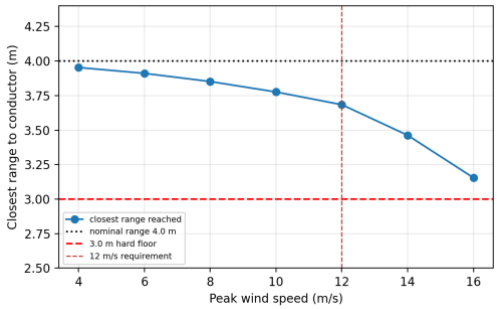


Figure 4. Closest range against peak wind speed.

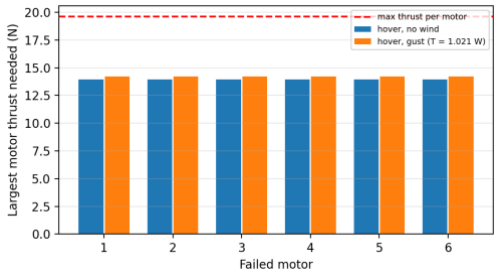


Figure 5. Largest motor thrust needed with each motor failed.

WHAT THE RESULTS SHOW:

- **Gusts:** in the 12 m/s gust we get pushed about 0.32 m toward the line and tilt about 11.7°. The closest range is 3.68 m, above the 3.0 m floor. A 0.5 s LiDAR dropout costs almost nothing. The sweep (Figure 4) stays above 3 m up to 16 m/s (3.15 m).
- **Thrust is not the limit:** peak thrust is 9.5 N per motor of 19.6 N. What decides whether we hold the range is the sensing and the controller tuning. With a weak velocity loop the same gust takes us to 2.57 m. We will tune against ArduPilot SITL before we claim anything.
- **Why 4.0 m and not 3.0 m:** if we flew exactly 3.0 m, even a 0.3 m push toward the line would break the stand-off. The 1.0 m margin is what makes the result pass.
- **Motor-out:** the largest thrust needed is 13.9 N in hover (71 % of 19.6 N) and 14.2 N in the gust (73 %), with zero residual roll, pitch and yaw moment. Torque-to-thrust ratio is assumed as 0.02 m.
- **Arms:** at max thrust the CFRP root stress is 11.0 MPa and the GFRP stress at the joint is 6.0 MPa. With a 1.5 safety factor the tip deflection is 1.4 mm. Against our assumed allowables (200 and 150 MPa) the factors are above 12, so the arms look over-built and a lighter tube may be possible. The first bending frequency is about 44 Hz, which we have to compare with the rotor frequency at hover.

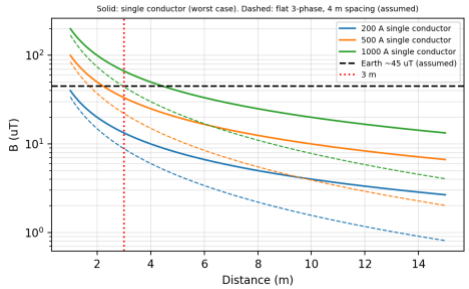


Figure 6. Magnetic field against distance. Solid: single conductor (worst case). Dashed: flat three-phase line with 4 m spacing (assumed).

LIMITS OF THESE RESULTS:

- The wind is uniform and always along the approach axis. Real gusts have side components, and the lattice of a tower makes its own wake. We have not modelled that.

- Rotor and ESC dynamics are collapsed into one attitude response, sensor noise is simple Gaussian noise, and altitude hold is ideal. Battery sag, cold and hot-day density are not modelled.
- Mass, efficiency, drag area, arm stiffness and allowable stresses are assumptions. They must be replaced with CAD, datasheet and test values. These results support the concept, they are not flight data.